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Division algorithm in $\mathbb{Z}[\sqrt{d}]$ for $d = -1, \pm 2, 3$.

Thm (DA in $\mathbb{Z}[\sqrt{d}]$) Given $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$ with $\beta \neq 0$,

there are $r, s \in \mathbb{Z}[\sqrt{d}]$ s.t.

$$\alpha = \beta r + s, \quad |N(s)| < |N(\beta)|.$$

Prop ($\mathbb{Z}[\sqrt{d}]$ is a PID) Every ideal I of $\mathbb{Z}[\sqrt{d}]$ is of

the form $I = (\alpha) = \{\alpha\beta : \beta \in \mathbb{Z}[\sqrt{d}]\}$.

Defn (Greatest common divisor) Let $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$, not both

0. A gcd of α, β is some $\delta \in \mathbb{Z}[\sqrt{d}]$ s.t.

① $\delta \mid \alpha$, $\delta \mid \beta$.

② If $\varepsilon \mid \alpha$, $\varepsilon \mid \beta$, then $\varepsilon \mid \delta$.

Lemma (Bézout) Let $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$, not both 0, where

$d \in \{-1, \pm 2, 3\}$. Let $\delta = (\alpha, \beta)$. Then the set

$$I_{\alpha, \beta} = \{\alpha r + \beta \varepsilon : r, \varepsilon \in \mathbb{Z}[\sqrt{d}]\}$$

is an ideal and $I_{\alpha, \beta} = (\delta)$. In particular, there are

$r, \varepsilon \in \mathbb{Z}[\sqrt{d}]$ s.t.

$$\delta = \alpha r + \beta \varepsilon.$$

$$\text{pf. } \delta/\alpha, \delta/\beta \Rightarrow \delta/\alpha r + \beta \varepsilon, \forall r, \varepsilon \in \mathbb{Z}[\sqrt{d}]$$

$$\Rightarrow I_{\alpha, \beta} \subseteq (\delta).$$

To show $(\delta) \subseteq I_{\alpha, \beta}$, we use the fact that $I_{\alpha, \beta}$ is principal, so $I_{\alpha, \beta} = (\delta')$. It suffices to show δ'/δ .

$$\text{When } r=1, \varepsilon=0, \alpha r + \beta \varepsilon = \alpha \in I_{\alpha, \beta} = (\delta'),$$

$$\text{When } r=0, \varepsilon=1, \alpha r + \beta \varepsilon = \beta \in I_{\alpha, \beta} = (\delta').$$

So δ'/α and δ'/β . Since $\delta = \gcd(\alpha, \beta)$, we have

$$\delta'/\delta.$$



Lem (Enclid) Let $d \in \{-1, \pm 2, 3\}$. If $\pi \in \mathbb{Z}[\sqrt{d}]$ is prime

and $\pi/\alpha\beta$ for some $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$, then π/α or π/β .

Observation: If $\pi \nmid \alpha$, then $(\pi, \alpha) = 1$.

Pf. Let $\delta \mid \pi$, $\delta \mid \alpha$. Since π is prime, δ is a unit or an associate of π . Since $\pi \nmid \alpha$, δ is a unit.

Pf of Euclid's lemma: $\pi \nmid \alpha$ ✓

So assume $\pi \nmid \alpha$ and show $\pi \mid \beta$.

By Observation, $(\pi, \alpha) = 1$.

Bézout $\Rightarrow$ there are $r, e \in \mathbb{Z}[\sqrt{\alpha}]$ s.t.

$$\pi r + \alpha e = 1.$$

$$\Rightarrow \pi \mid \pi r + \alpha \mid \beta e = \beta$$

$$\pi | \pi, \pi | \alpha\beta \Rightarrow \pi | \pi\beta r + \alpha\beta e = \beta.$$



Thm (Unique factorization in $\mathbb{Z}[\sqrt{d}]$, $d = -1, \pm 2, 3$)

Every nonzero, nonunit element of $\mathbb{Z}[\sqrt{d}]$ can be expressed as a product of primes, uniquely up to order and multiplication by units.

Pf. Existence ✓

Uniqueness: ① α is prime: ✓

② α is composite:

Assume α is a composite element with minimal $|N(\alpha)|$

and two distinct prime factorizations:

$$\alpha = \pi_1 \pi_2 \cdots \pi_k = \rho_1 \rho_2 \cdots \rho_\ell \quad (k, \ell \geq 2) \quad (*)$$

$$\pi_1 \mid \rho_1 \cdots \rho_\ell \xrightarrow{\text{Euclid}} \pi_1 \mid \rho_j \text{ for some } 1 \leq j \leq \ell.$$

Rearranging $\rho_1, \dots, \rho_\ell$, we may assume $j=1$.

So $\pi_1 \mid \rho_1$. But π_1, ρ_1 are prime. So $\rho_1 = \varepsilon \pi_1$ for some

unit ε . Substituting this into $(*)$,

$$\alpha = \pi_1 \pi_2 \cdots \pi_k = \varepsilon \pi_1 \rho_2 \cdots \rho_\ell.$$

$$\Rightarrow \frac{\alpha}{\pi_1} = \pi_2 \cdots \pi_k = \varepsilon \rho_2 \cdots \rho_\ell.$$

These are two distinct prime factorizations of the nonunit

$\beta = \frac{\alpha}{\pi_1}$. Since $\alpha = \pi_1 \beta$ implies

$$|N(\alpha)| = \underbrace{|N(\pi_1)|}_{>1} |N(\beta)| > |N(\beta)|.$$

this contradicts minimality of α .



Ex. $\mathbb{Z}[\sqrt{3}]$: $4 = 2 \cdot 2 = (1+\sqrt{3})(1-\sqrt{3})$



$$2 \mid 4 = (1+\sqrt{3})(1-\sqrt{3}) \text{ , but}$$

$$2 \nmid (1 \pm \sqrt{3}) .$$

So 2 does not satisfy Euclid's lemma.